

Lecture 10

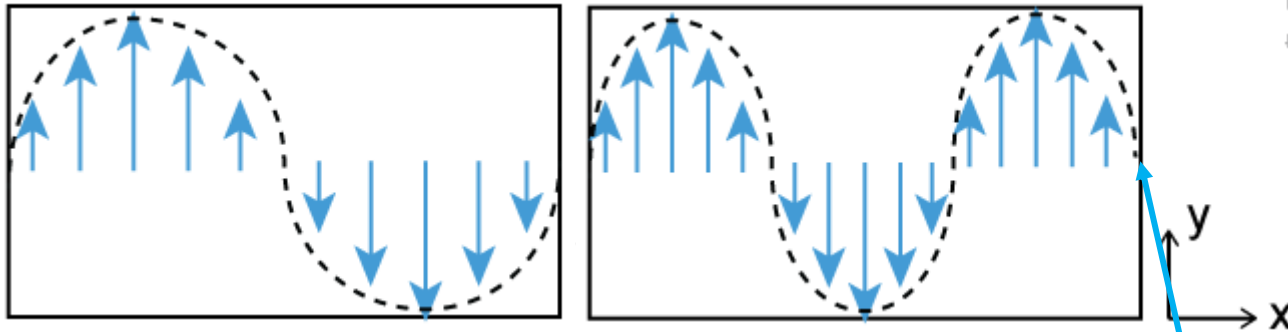
Monday 7/20/26

Sami Lewis

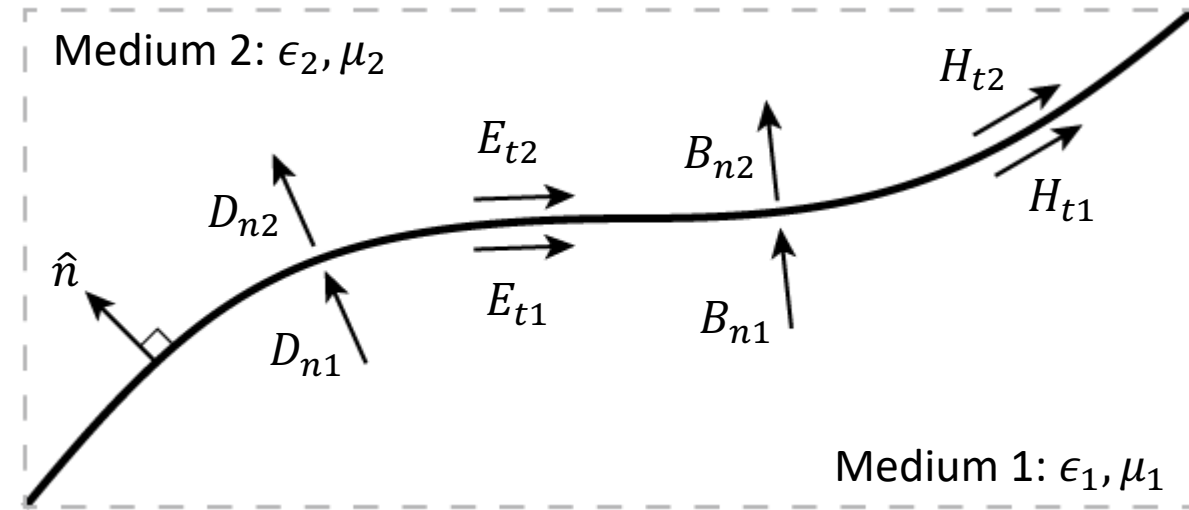
US Particle Accelerator School
Accelerator Physics – Summer 2026

Guided waves

Solutions are eigenmodes with a discrete number of peaks/nodes in x and y enforced by the boundary conditions of the walls.
Rectangular E field example:



Oscillates with phase, fixed number of nodes



Fields cannot exist inside a perfect conductor, gives boundary conditions for metallic walls:

$$\hat{n} \cdot \mathbf{D} = \rho_s$$

$$\hat{n} \times \mathbf{E} = 0$$

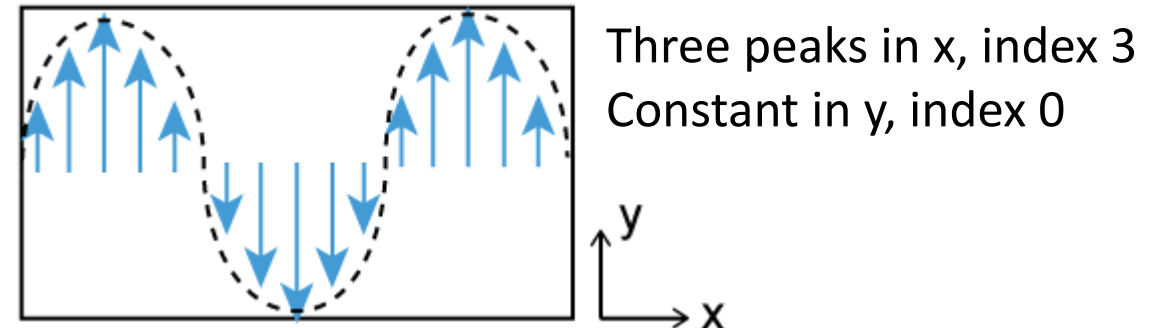
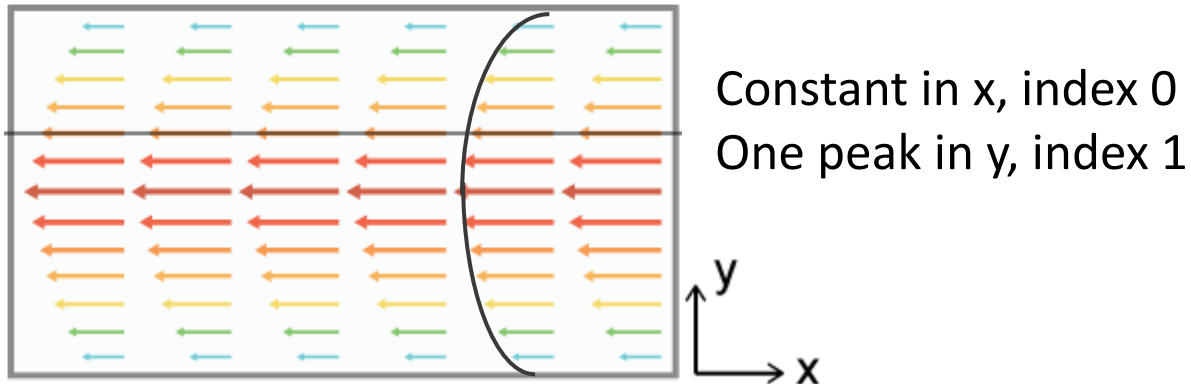
$$\hat{n} \cdot \mathbf{B} = 0$$

$$\hat{n} \times \mathbf{H} = \mathbf{J}_s$$

Tangential E field must go to zero at the walls

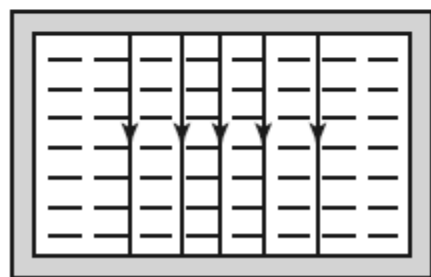
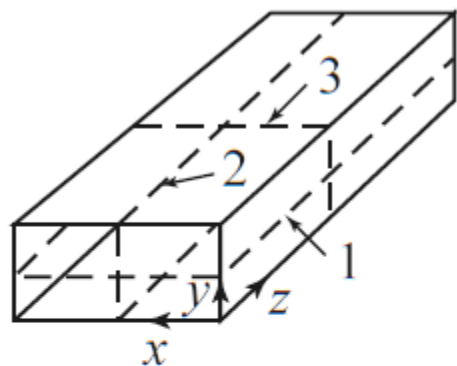
Types of modes

- Mode naming:
 - $E_z = 0$ “Transverse Electric” or **TE modes**, have non-zero H_z
 - $H_z = 0$ “Transverse Magnetic” or **TM modes**, have non-zero E_z
 - $E_z = H_z = 0$ “Transverse Electromagnetic” or **TEM modes**, requires two conductors
- Modes are named with indices specifying the number of peaks/nodes in each direction, e.g. TM_{mn}
- My opinion: easiest to determine indices based on E field, count variations along x at fixed y, etc.

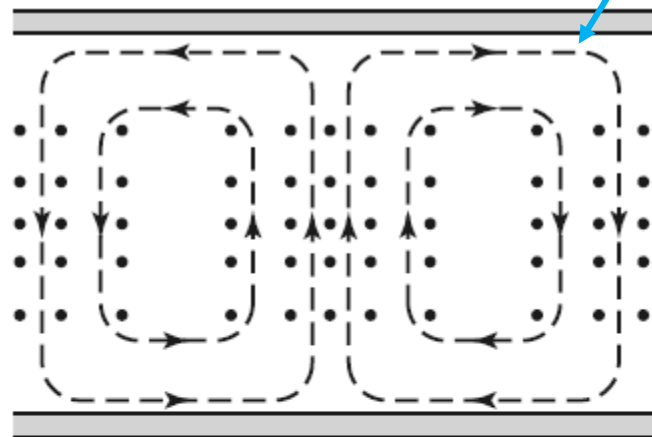


Examples: rectangular waveguide

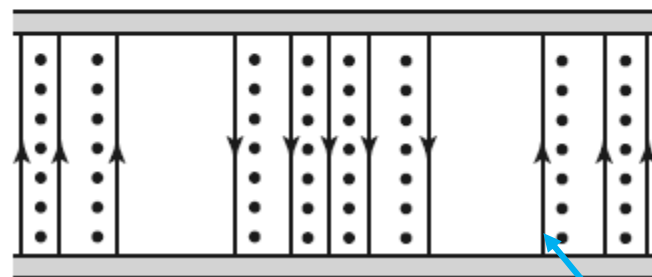
TE₁₀ mode



x-y plane 3

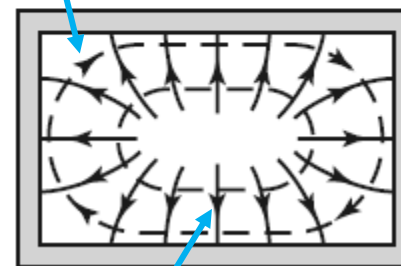
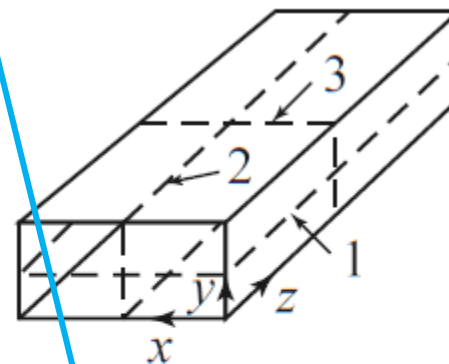


x-z plane 1



y-z plane 2

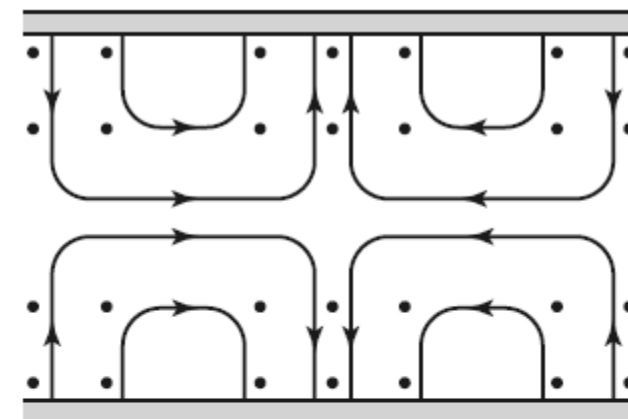
Dashed lines: H field



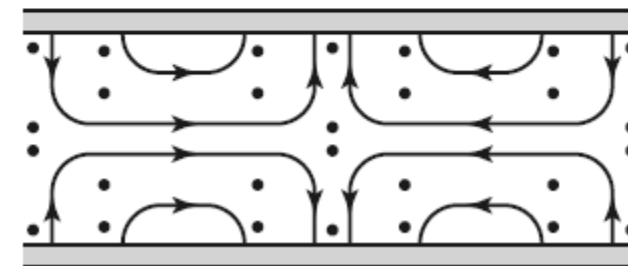
x-y plane 3

Solid lines: E field

TM₁₁ mode



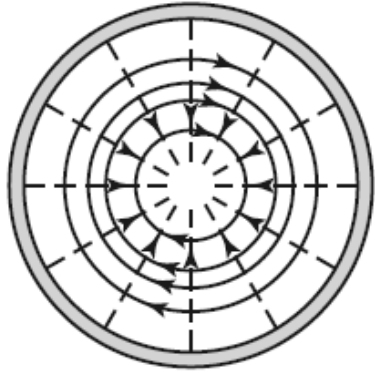
x-z plane 1



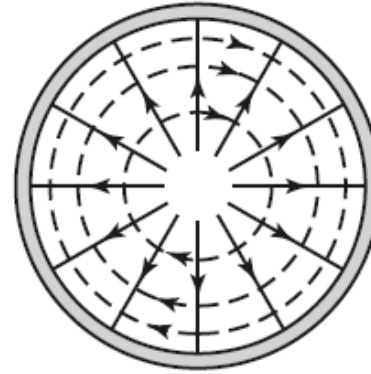
y-z plane 2

Examples: circular waveguide

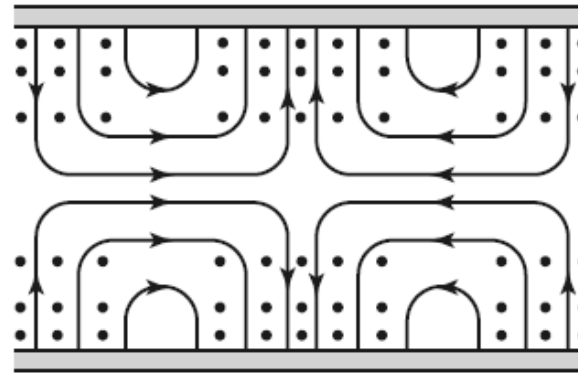
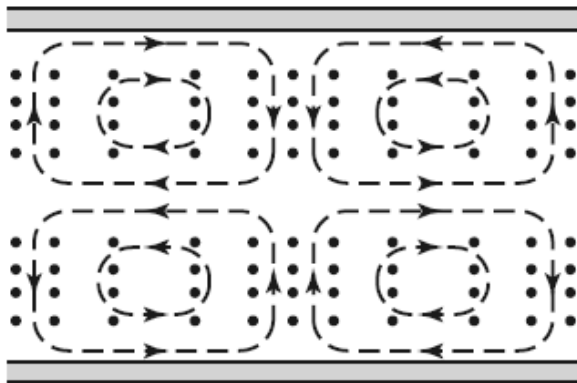
TE_{01}



TM_{01}



TM modes: our accelerating modes (generally), have axial component of E field



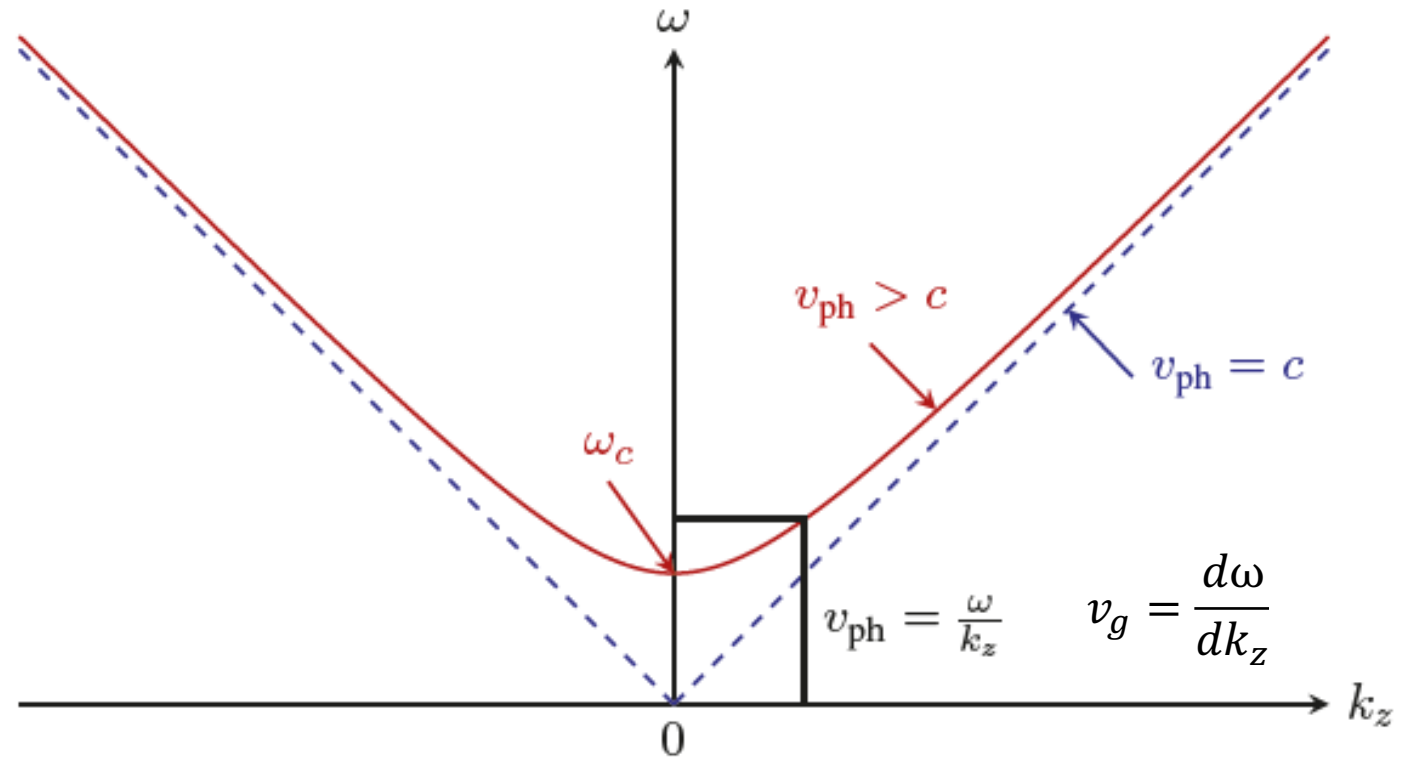
Dispersion relation

$$k_z^2 = k^2 - k_c^2$$

$$k_c = \frac{2\pi}{\lambda_c} = \frac{\omega_c}{c}$$

Cutoff frequency

$$k_z^2 = \frac{\omega^2}{v_p^2} = \frac{\omega^2 - \omega_c^2}{c^2}$$



<https://cds.cern.ch/record/1401088/files/p277.pdf>

Velocities in waveguides

$$v_p = \frac{\omega}{k_z}$$

Phase velocity: speed of phase fronts, possible to be $> c$

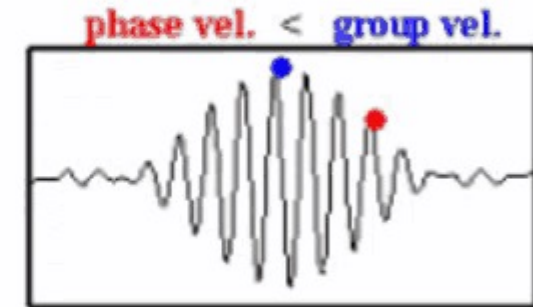
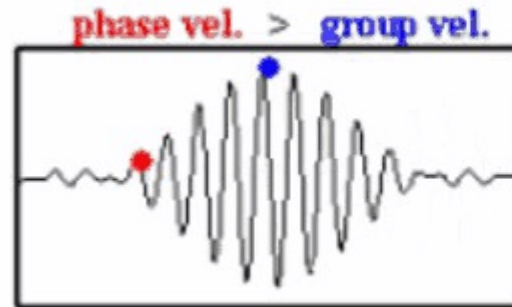
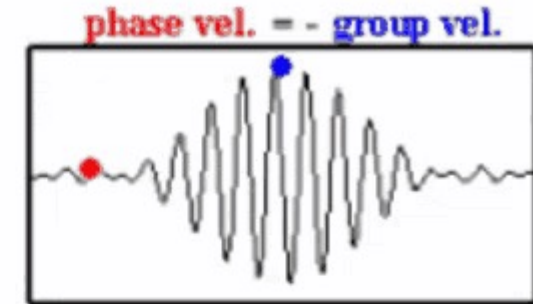
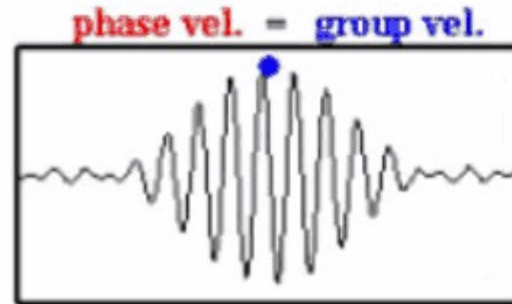
$$v_g = \frac{d\omega}{dk_z}$$

Group velocity: speed of energy flow, never $> c$

Velocities in waveguides

$$v_p = \frac{\omega}{k_z}$$

$$v_g = \frac{d\omega}{dk_z}$$



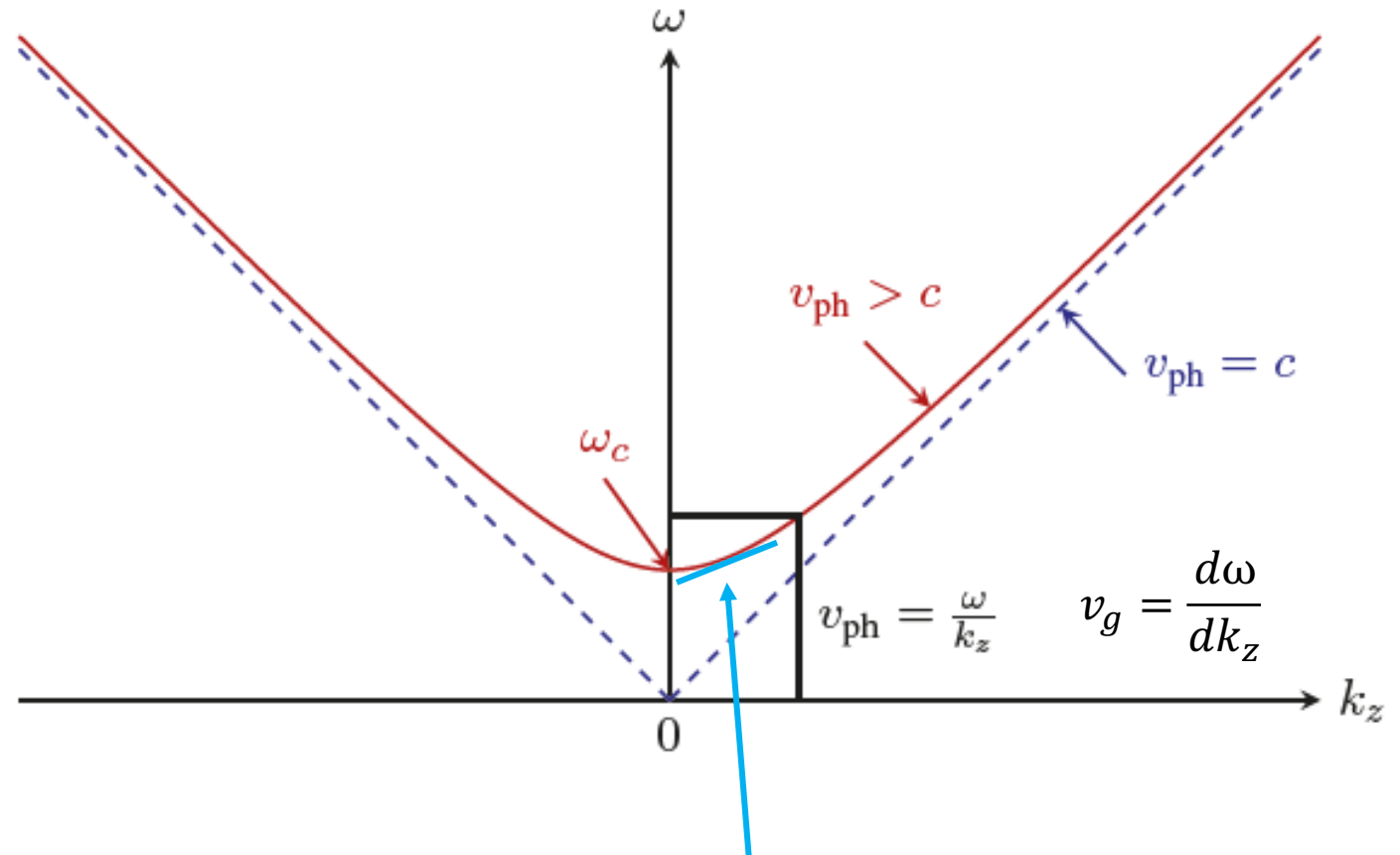
Dispersion relation

$$k_z^2 = k^2 - k_c^2$$

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Cutoff frequency

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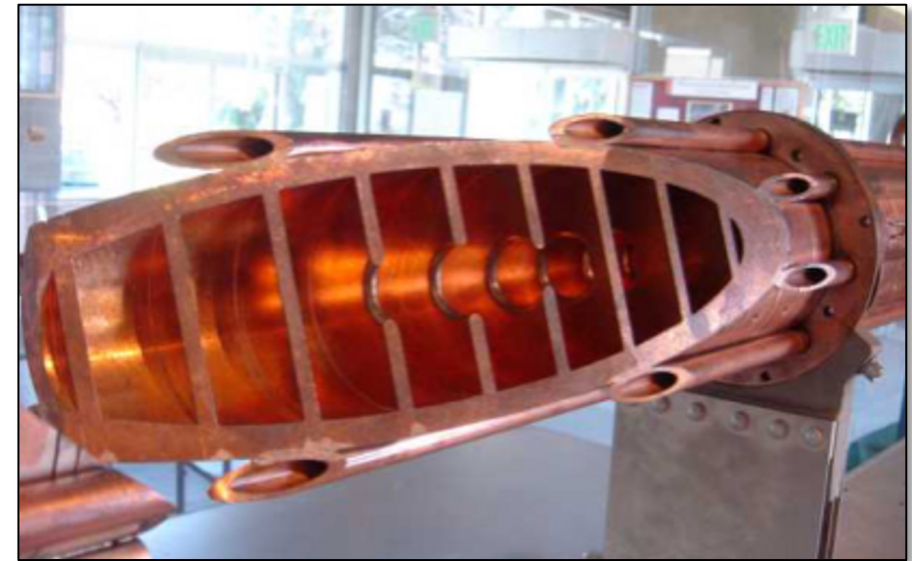
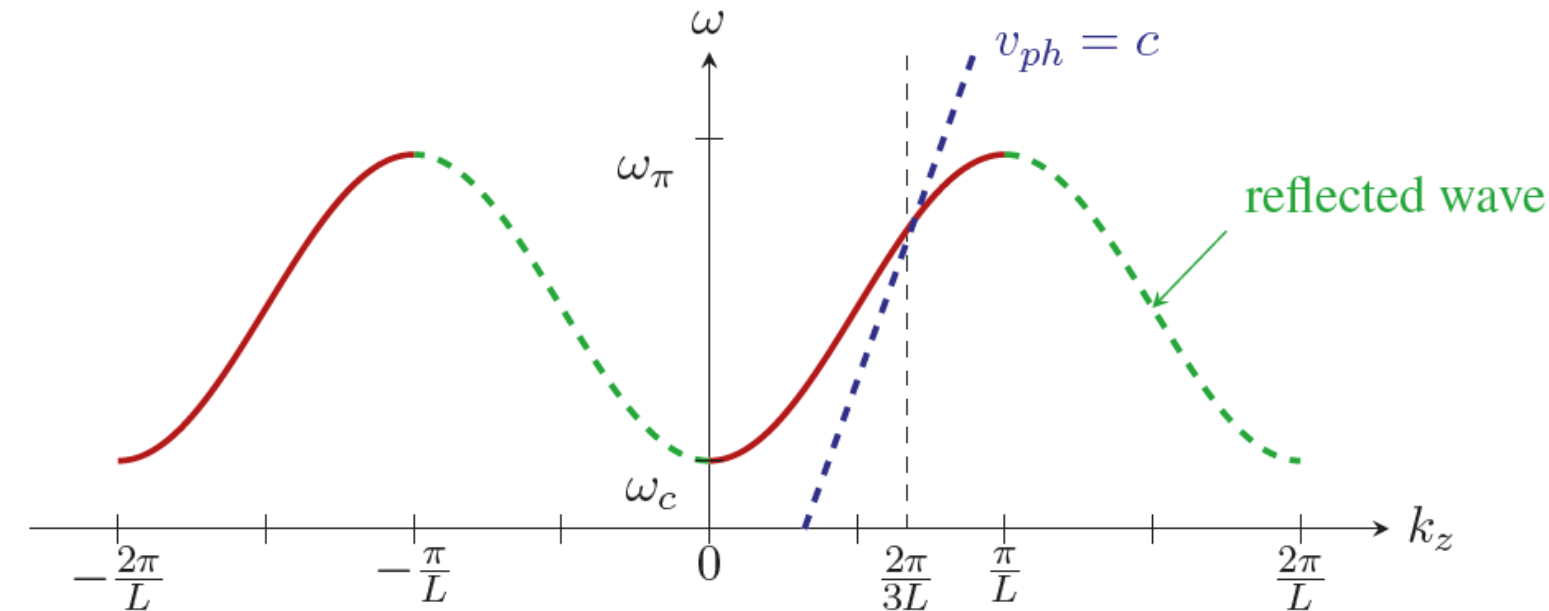
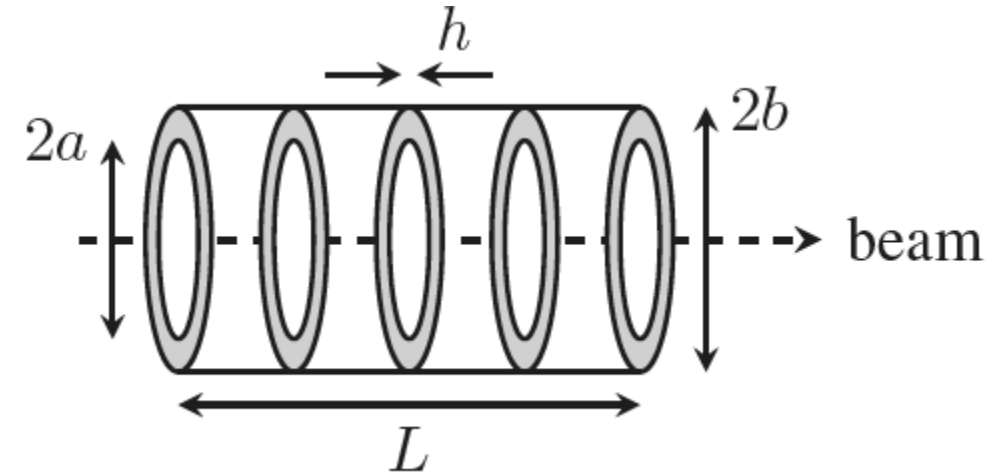


Slope: group velocity, never $> c$

<https://cds.cern.ch/record/1401088/files/p277.pdf>

Slow wave structures

- Want to reduce phase velocity to match particle velocity for e.g. traveling-wave accelerator
- Change effective permittivity and permeability
- E.g. dielectric loading, disc loaded waveguide

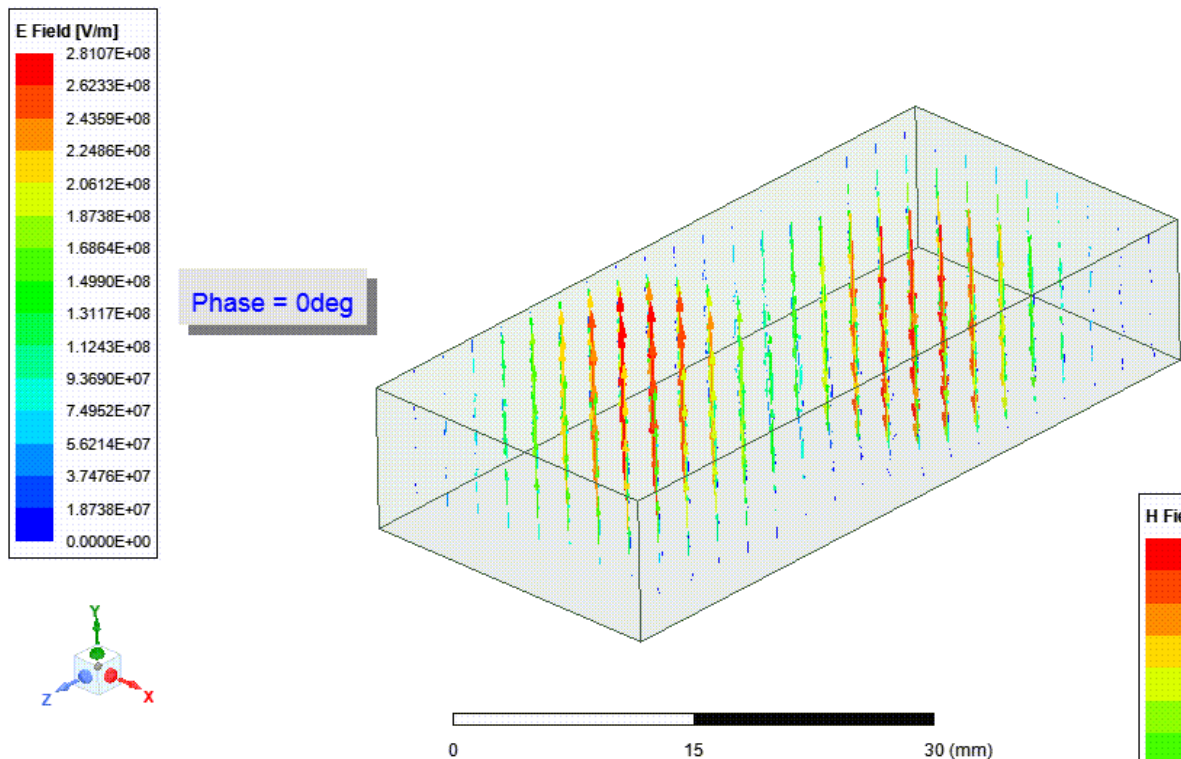


Cavities overview

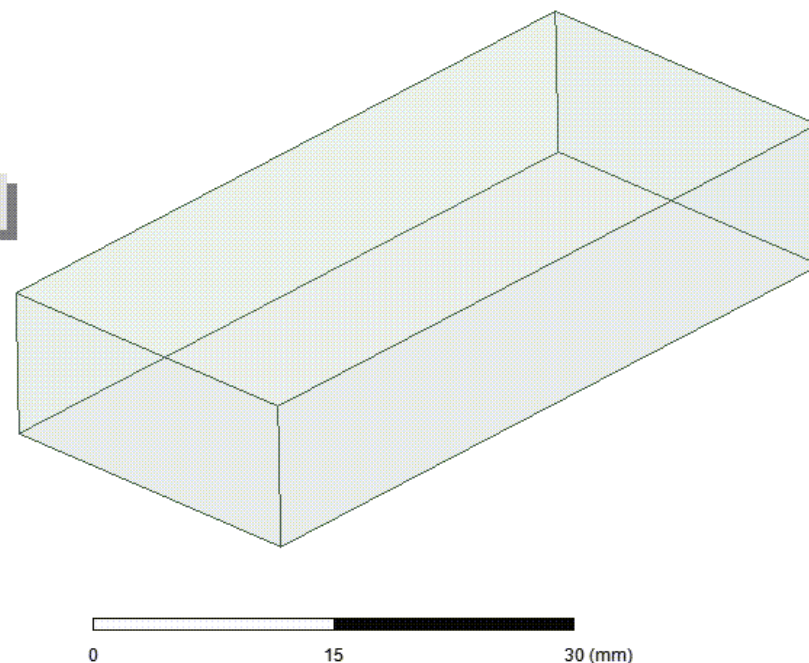
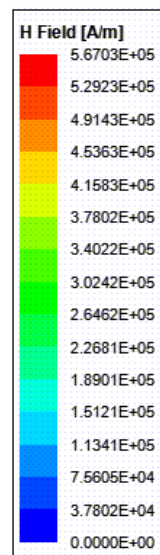
Resonant cavities

- Simplest picture: put reflective boundaries at each end of a waveguide, create a standing wave
- Now we enforce a number of nodes in the longitudinal (z) direction
- Mode types/names remain the same, but add an index: e.g. $\text{TM}_{mn\ell}$

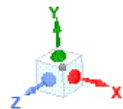
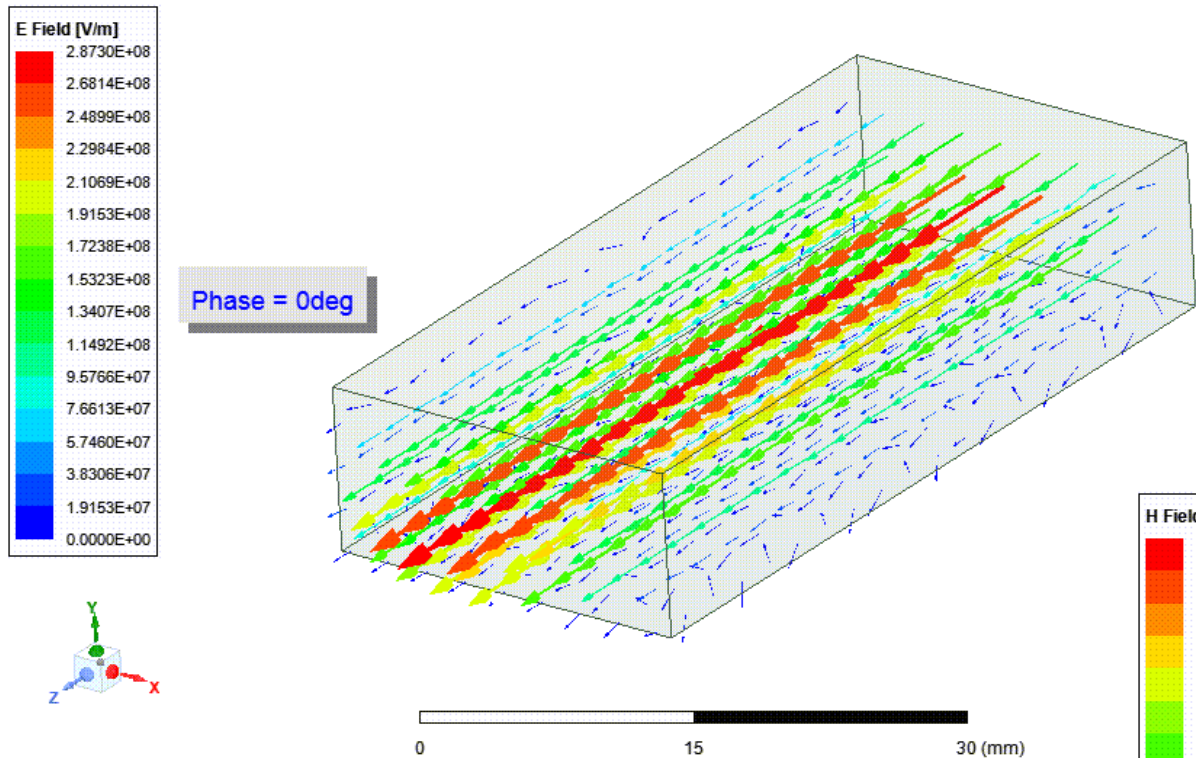
Rectangular example: TE mode



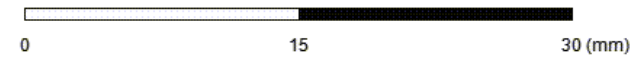
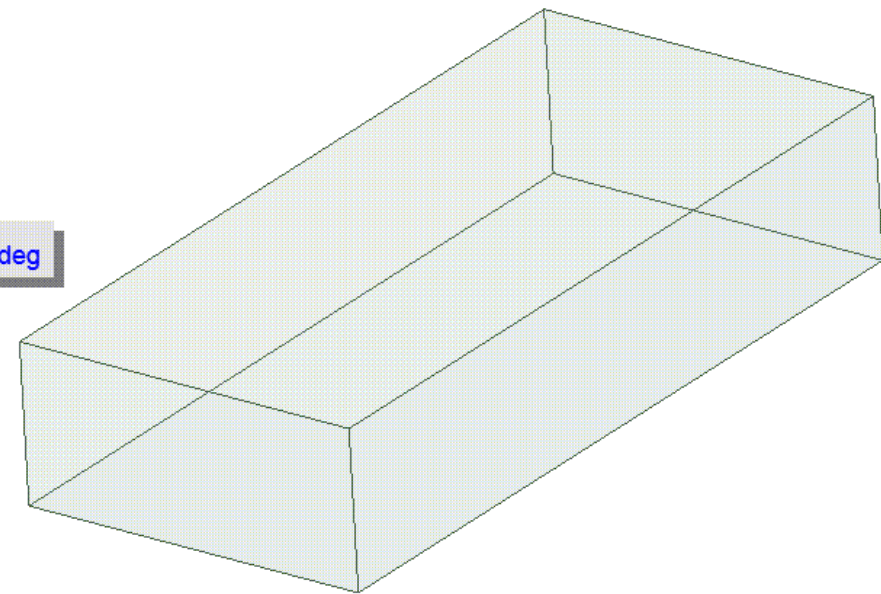
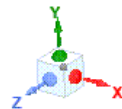
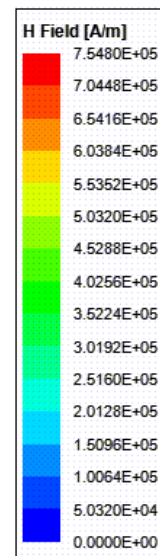
TE₁₀₂



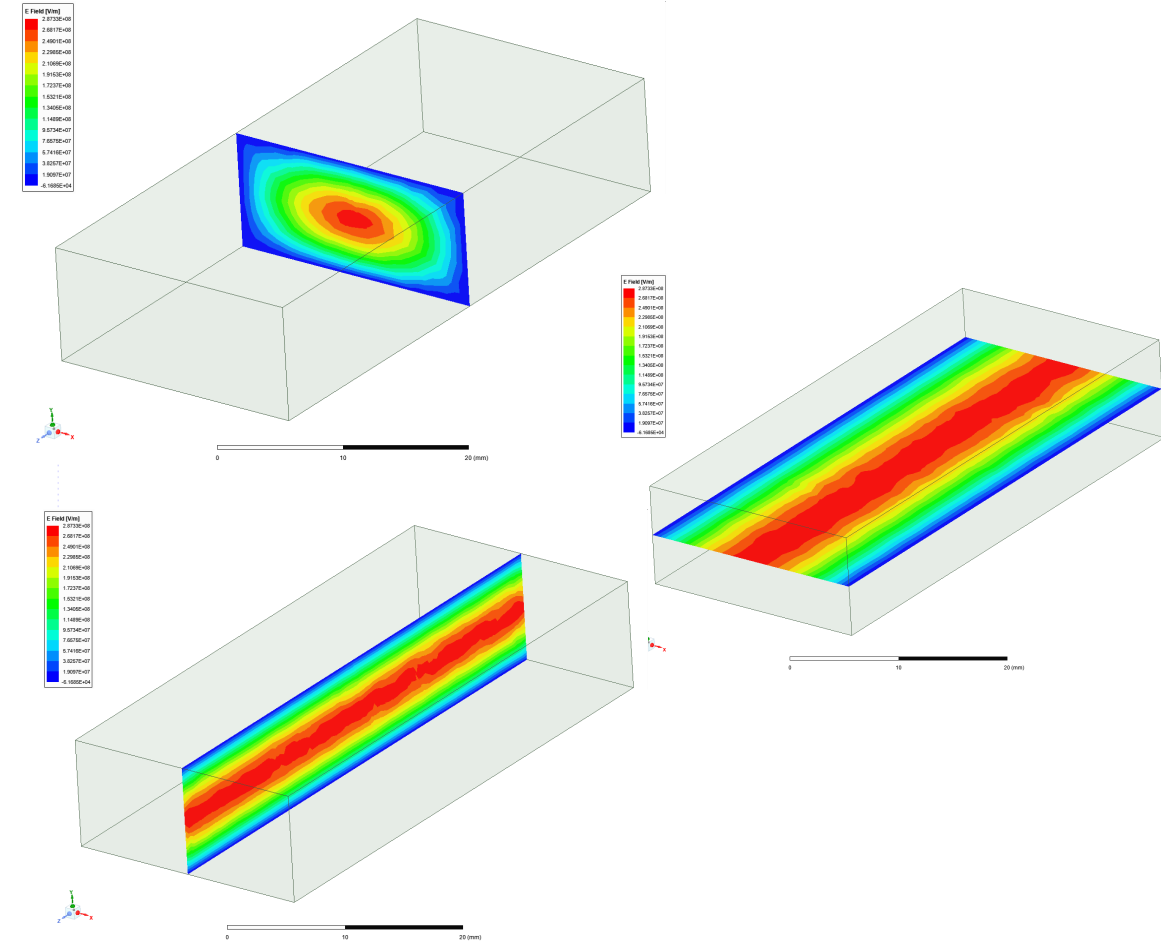
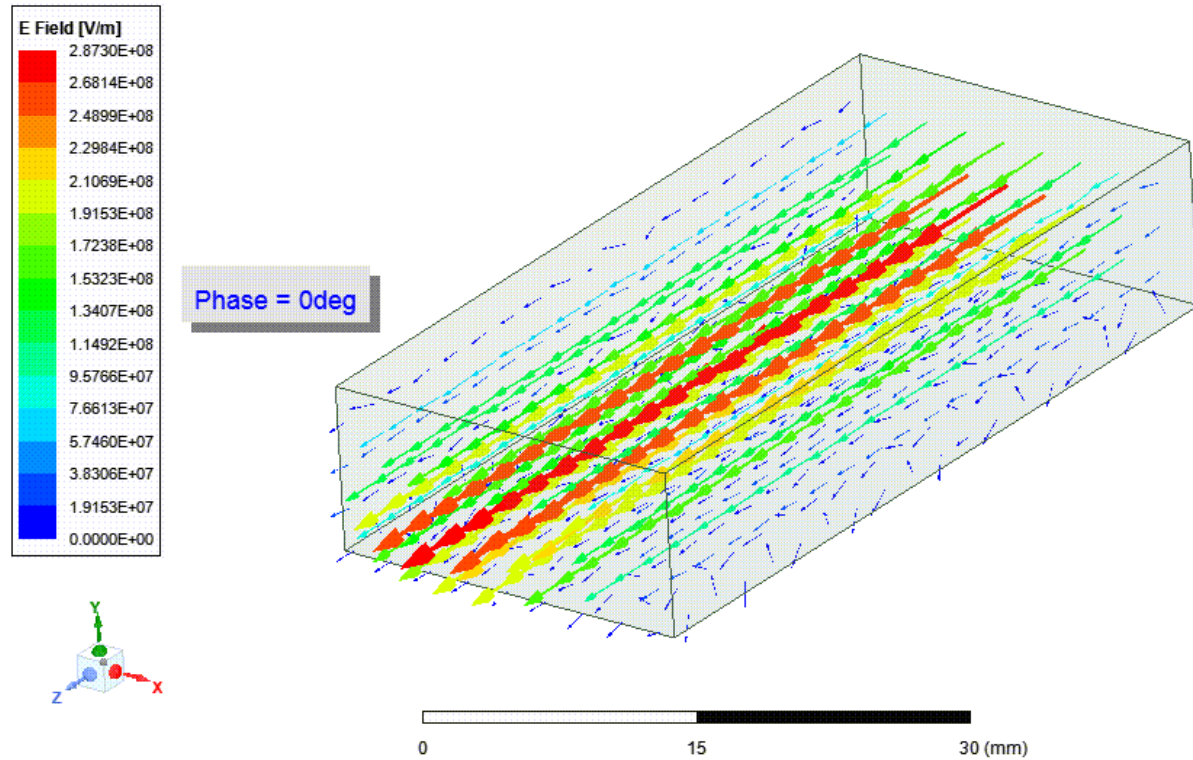
Rectangular example: TM mode



TM₁₁₀



Complex magnitude



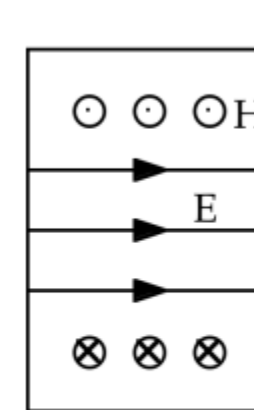
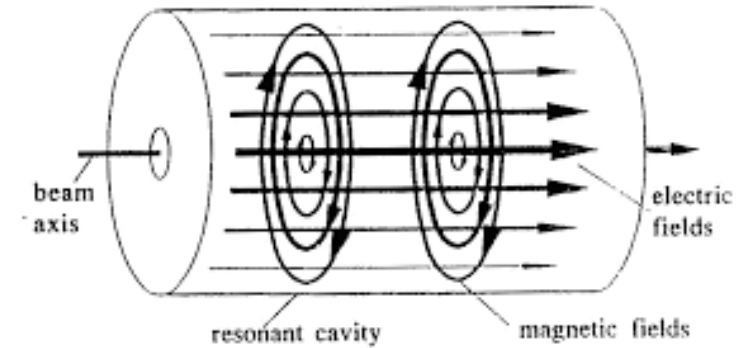
Pillbox cavity

- The frequencies of cylindrical cavity modes depend on the dimensions of the cavity and the Bessel function roots p_{mn} and p'_{mn} (e.g. p_{mn} is the n th root of J_m)
- For $TM_{mn\ell}$ modes in a cavity with radius a and length d , the frequencies are given by

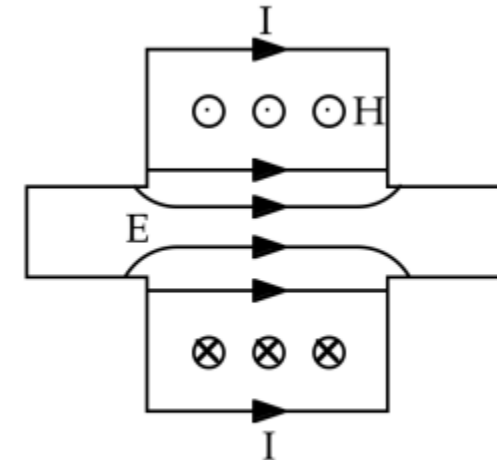
$$f_{mn\ell} = \frac{c}{2\pi\sqrt{\mu_r\epsilon_r}} \sqrt{\left(\frac{p_{mn}}{a}\right)^2 + \left(\frac{\ell\pi}{d}\right)^2}$$

- For acceleration, we (generally) use the TM_{010} mode. In vacuum:

$$f_{010} = \frac{2.405c}{2\pi a}$$



pillbox



pillbox + beam pipes

Cavity mode Q values

- Cavity modes are characterized by frequency and intrinsic Q value
- Q value is a measure of energy stored U vs power P lost per cycle
- For hollow metallic cavities, loss comes from surface resistance R_{surf} due to finite conductivity of the walls

$$Q_0 = \frac{\omega_0 U}{P}$$

Electric and magnetic stored energies:

$$U_e = \frac{\epsilon}{4} \int_V \mathbf{E} \cdot \mathbf{E}^* dV \quad U_m = \frac{\mu}{4} \int_V \mathbf{H} \cdot \mathbf{H}^* dV$$

Power lost through a good conductor:

$$P = \frac{R_{surf}}{2} \int_S |H_t|^2 dS \quad R_{surf} = \frac{1}{\sigma \delta_s}$$

“Skin depth” quantifies the penetration depth of fields into a material with high but finite conductivity (metals)

$$\delta_s = \sqrt{\frac{2}{\omega \mu \sigma}}$$

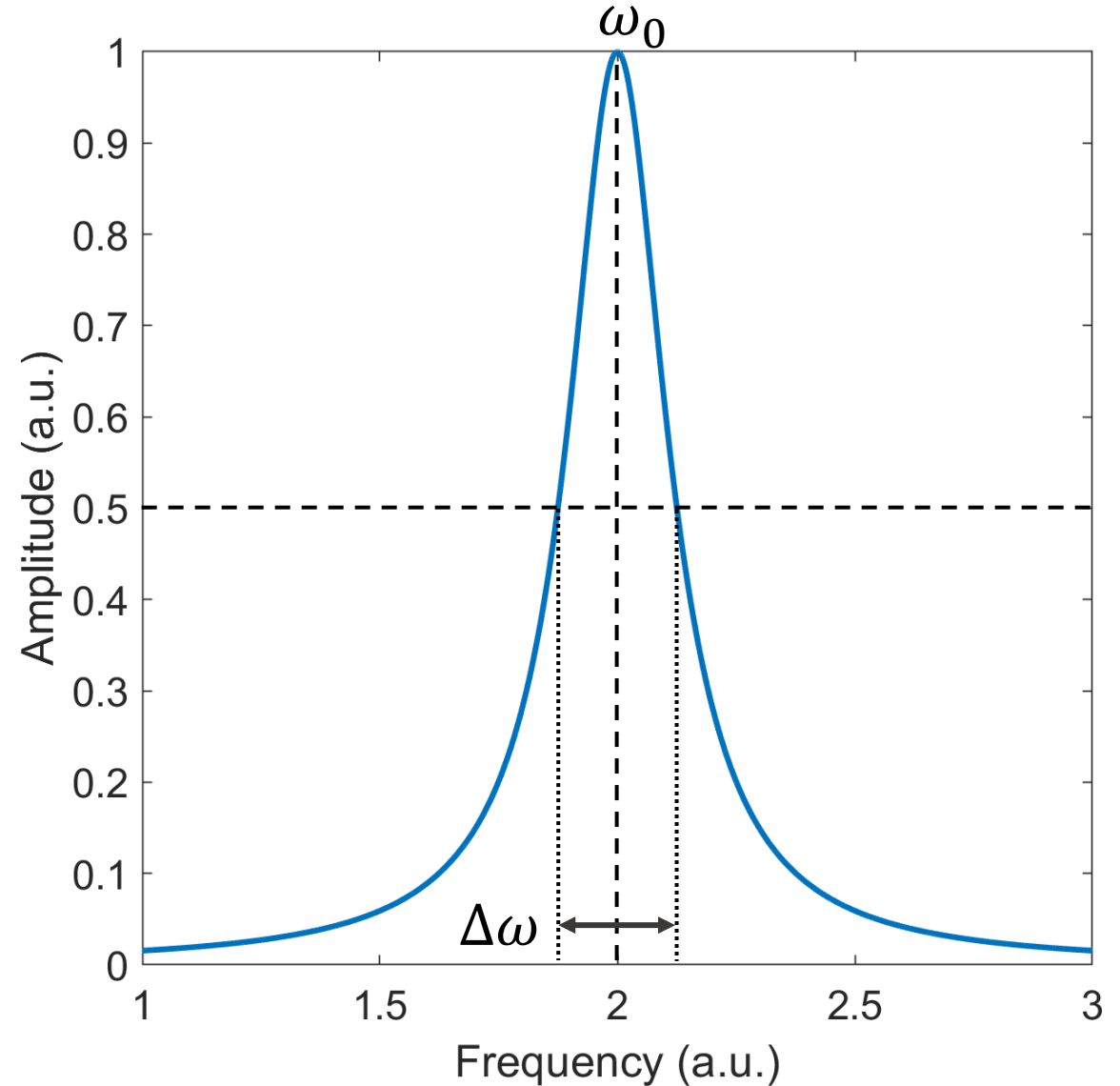
Cavity mode Q values

- Resonances have a Lorentzian lineshape
- We can determine Q_L from the bandwidth of the measured peak:

$$Q_L = \frac{\omega_0}{\Delta\omega}$$

- Higher Q value looks like a sharper peak
- Loaded Q value (Q_L) is related to intrinsic Q (Q_0) by coupling coefficient

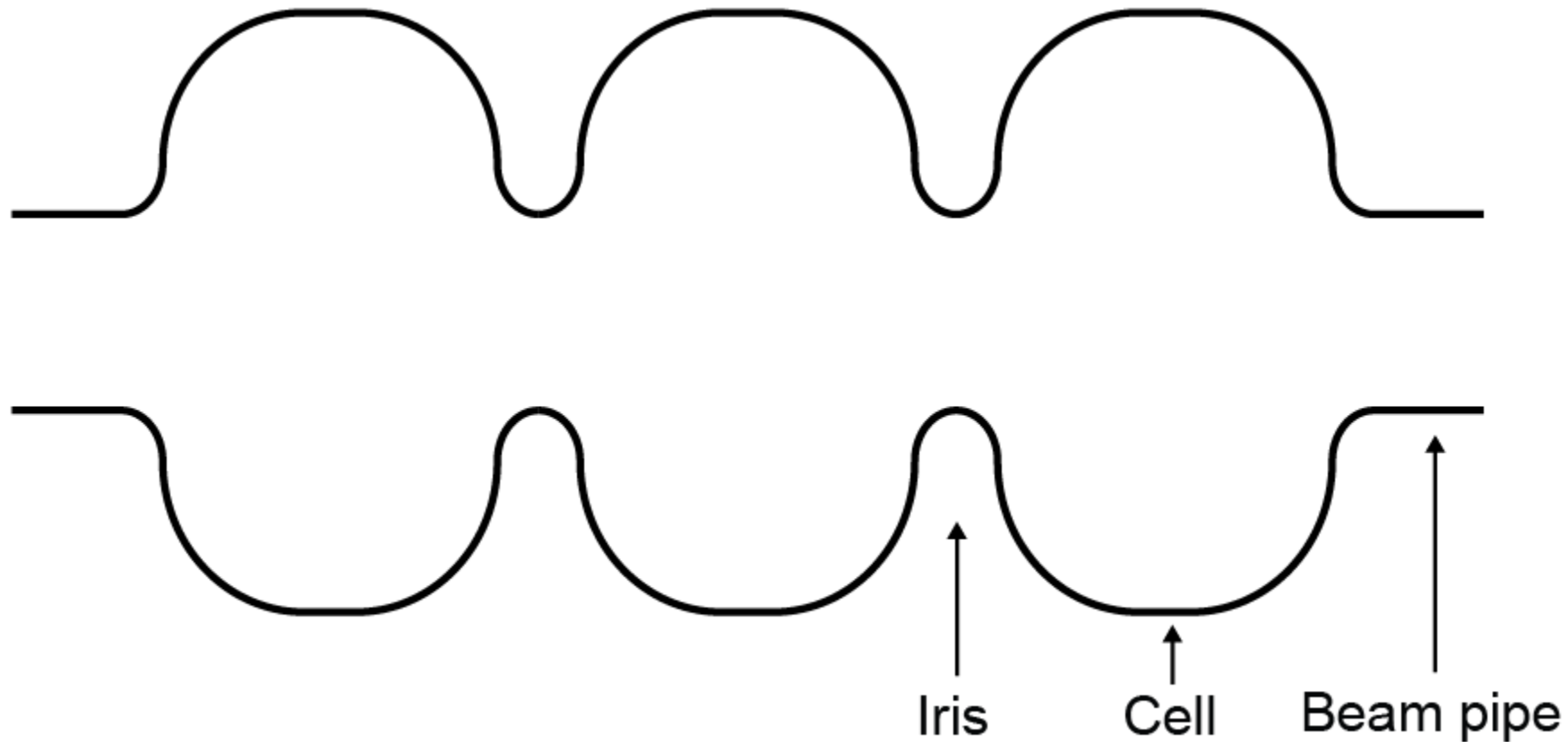
$$Q_L = \frac{Q_0}{1 + \beta_c}$$

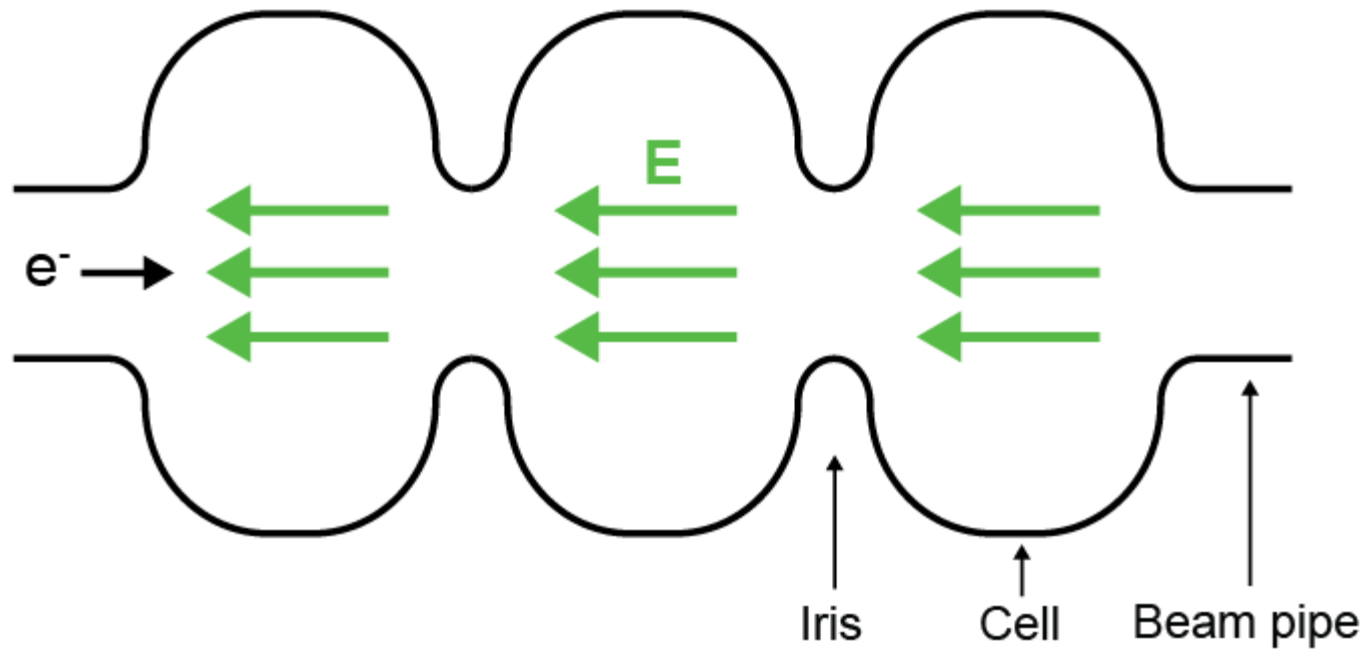


Accelerator cavities

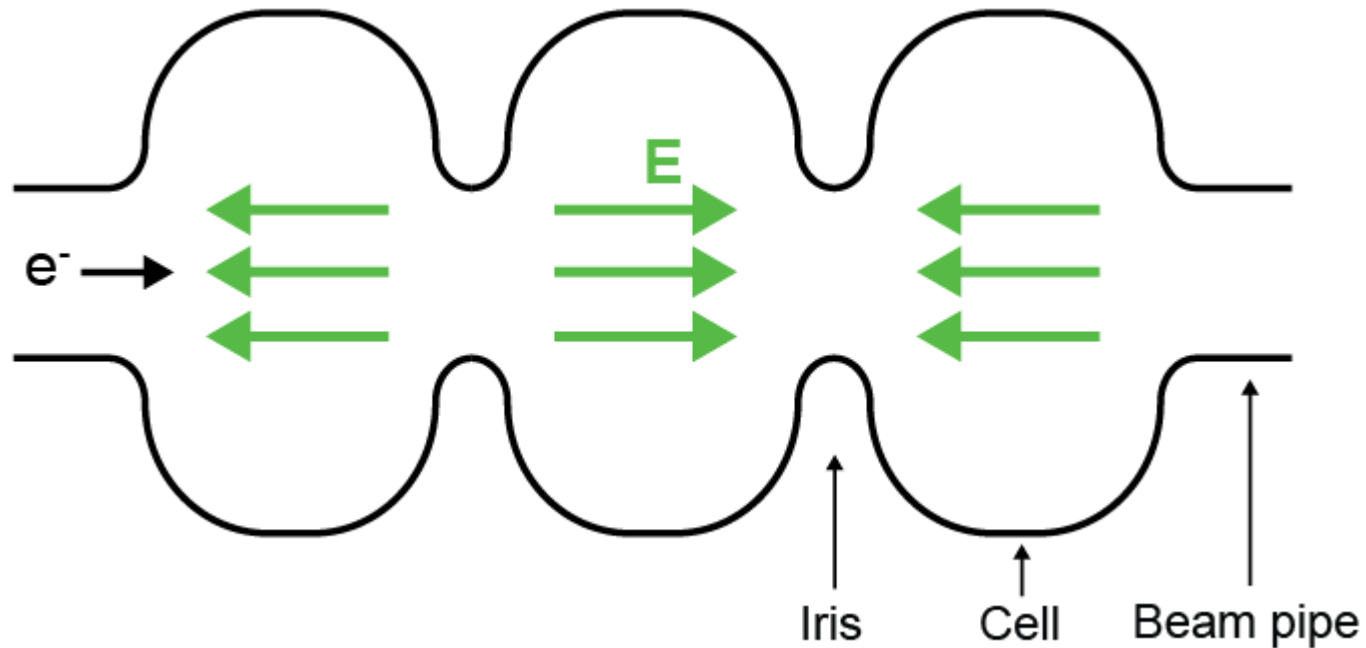
Standing wave accelerators

- Multi-cell cavities: coupled oscillators, multiple possible modes





0 mode
0 or 2π phase advance between cells

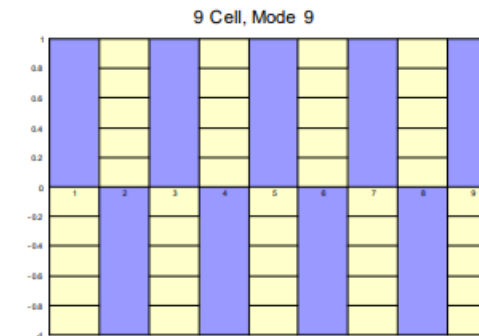
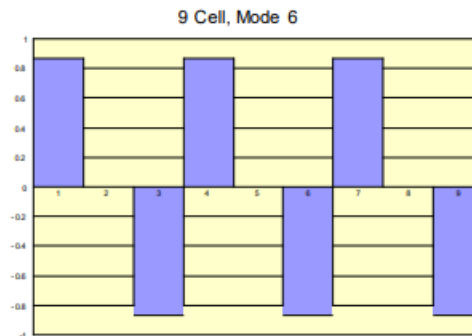
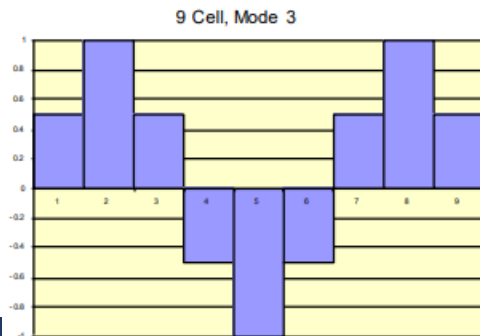
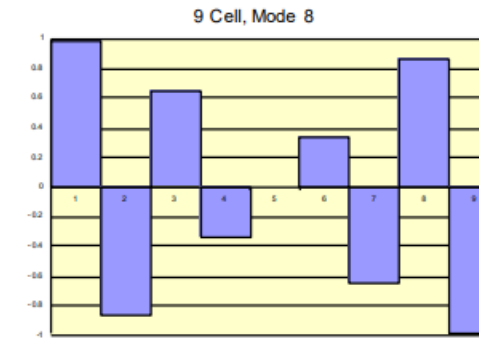
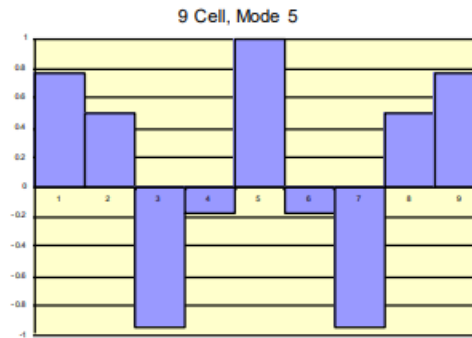
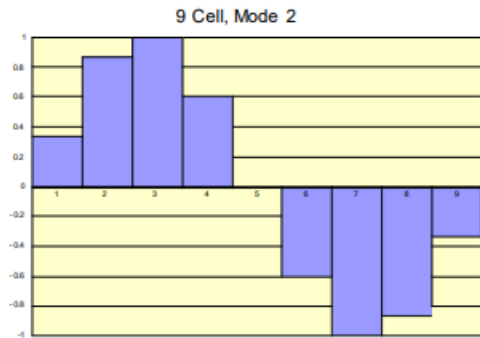
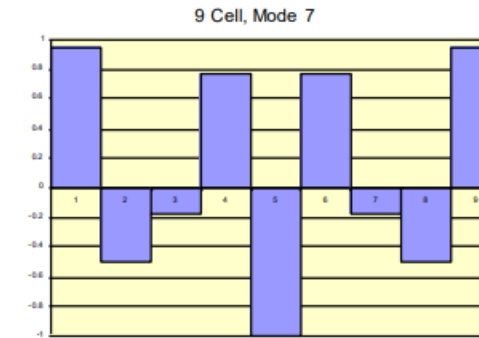
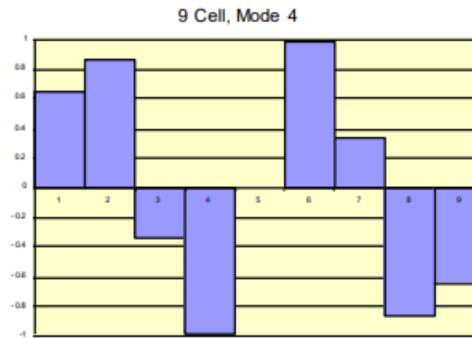
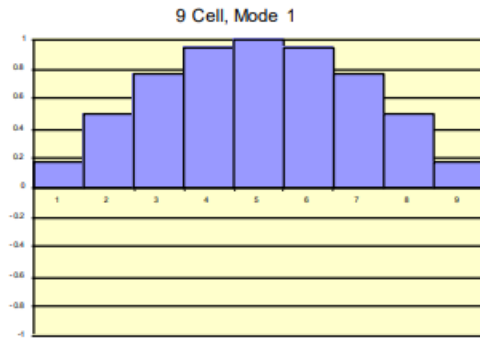


π mode
 π phase advance between cells

Multi-cell modes

- As many possible modes as there are cells, each has different frequency (9 cell example)

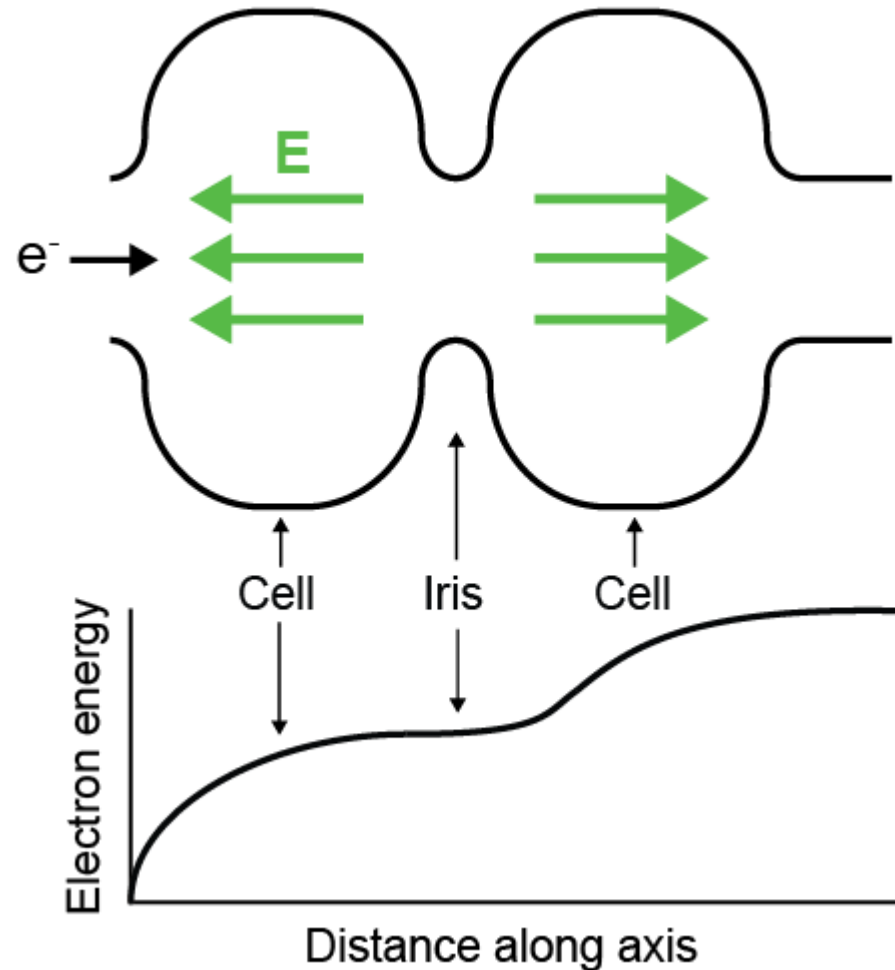
0 mode



π mode

Acceleration in multi-cell cavities

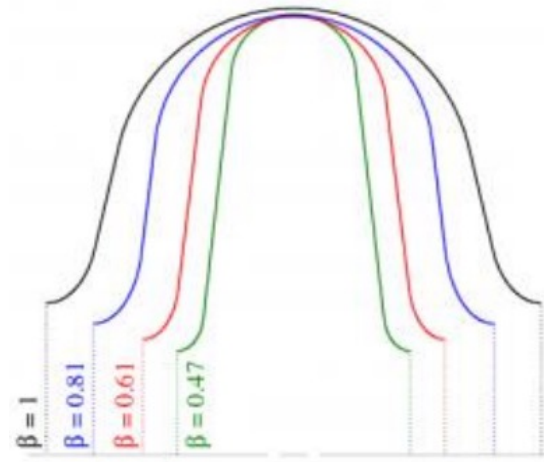
- Cavity length and frequency is designed such that particle is always accelerated (seeing correct phase of field)



- Recall:
$$f_{010} = \frac{2.405c}{2\pi a}$$
- Can have multiple lengths at the same frequency TM_{010} frequency
- Will affect e.g. 0 and π mode frequencies
- Limited set of RF sources needed

Cell length variation

- Example of different cavity lengths



SNS 805 MHz cavities ($\beta = 0.61$ and 0.81)

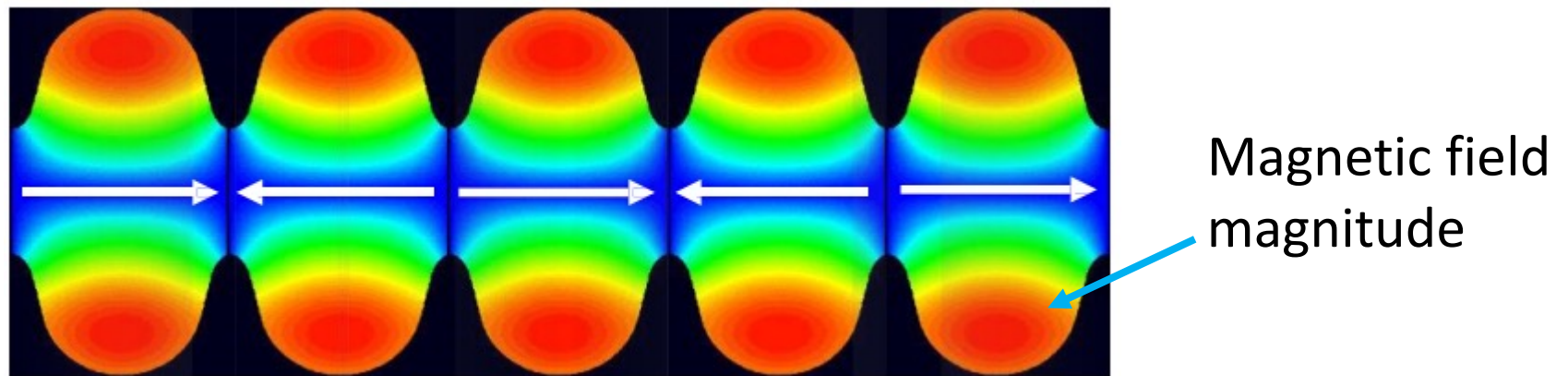
- Important for “low β ” where particle is still accelerating: sources and linacs

* Summer 2022 class all on SRF cavities: <https://sites.google.com/view/uspas2022-rf-superconductivity/home>

Important parameters

Limitations: pulsed heating

- Surface current causes heating of cavity walls (normal conducting structures)
- Heat diffusion happens at a finite rate, metal expands
- Source of stress and damage in material when heating is faster than expansion
- Over many RF cycles, can cause failure
- Thermal fatigue threshold: stress from each RF pulse is above the yield stress of the material
- Limits maximum possible fields, repetition rate of RF accelerators



Limitations: breakdown/field emission

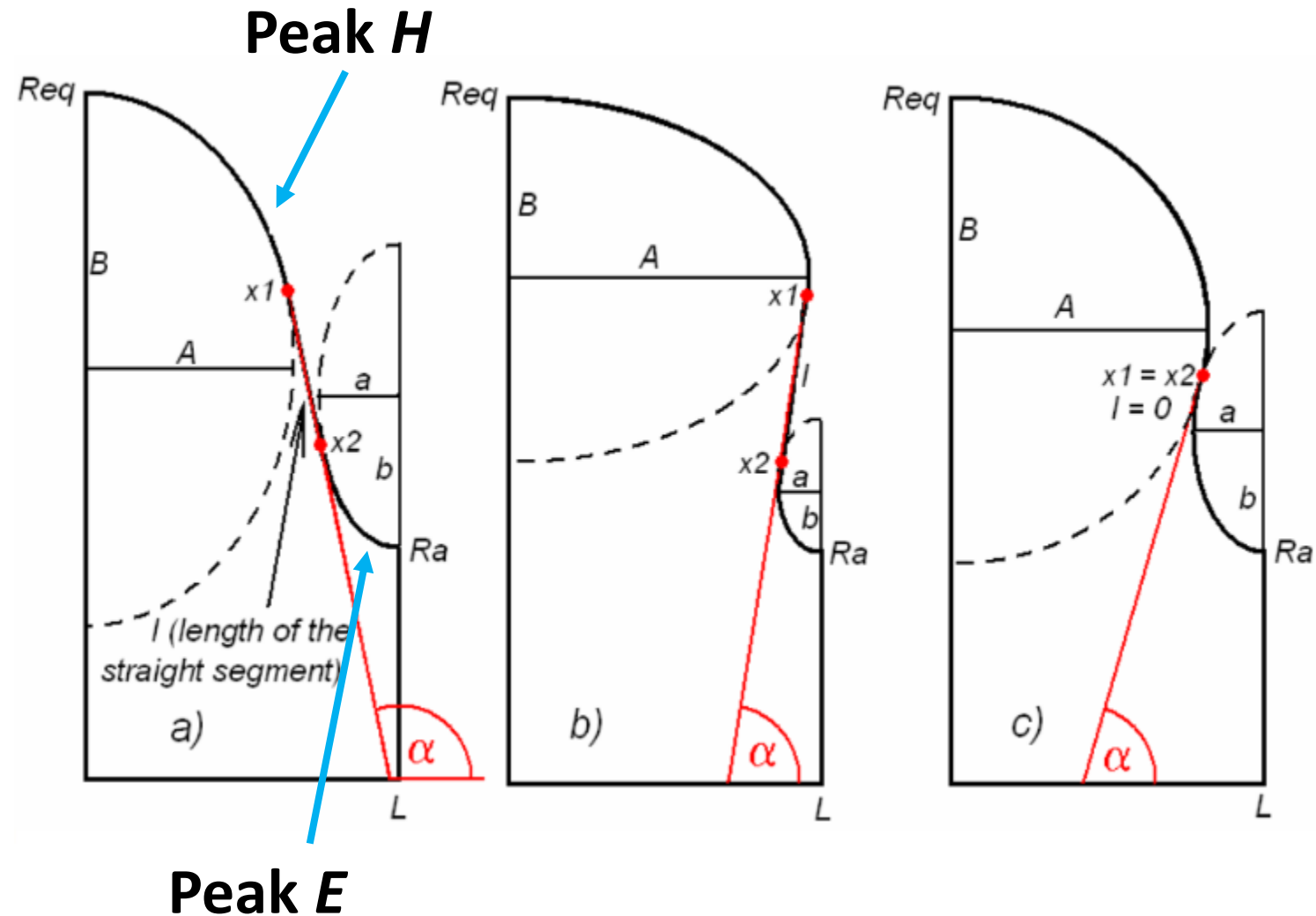
- Vacuum breakdown limits maximum achievable electric field and accelerating gradient
- Kilpatrick criterion/breakdown voltage (normal conducting cavities) with E_K in MV/m:

$$f[\text{MHz}] = 1.64 E_K^2 e^{-8.5/E_K}$$

- Rough empirical scaling with frequency, exceptions have been measured
- SRF case: talk about “field emission” instead
- SRF case: localized heating causes thermal breakdown, can lead to quenches

Cavity optimization

- Shape of elliptical-style cavities is highly optimized for specific cases
- Peak surface electric field on iris, peak surface magnetic field on cell wall
- Basic requirements: length and overall radius, sets frequency and possible particle energies
- Shape optimization: reduce pulsed heating, increase Q , increase shunt impedance, increase gradient, reduce potential spots for breakdown

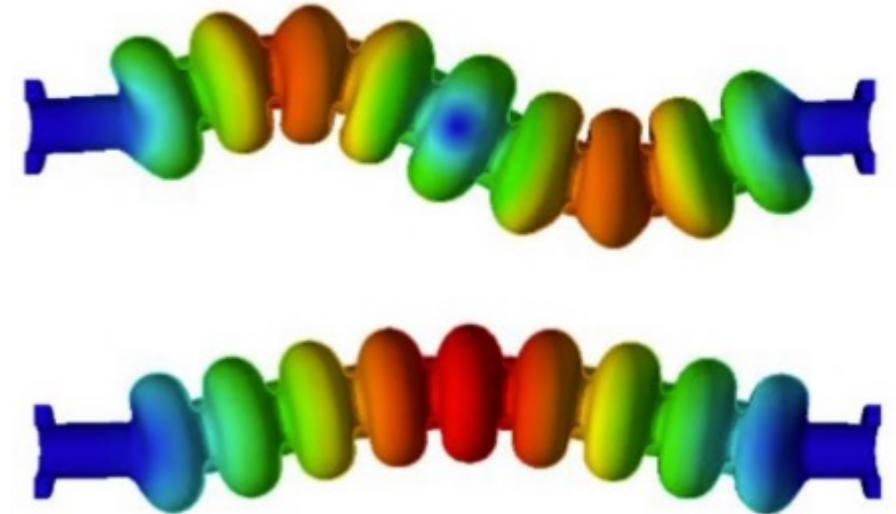
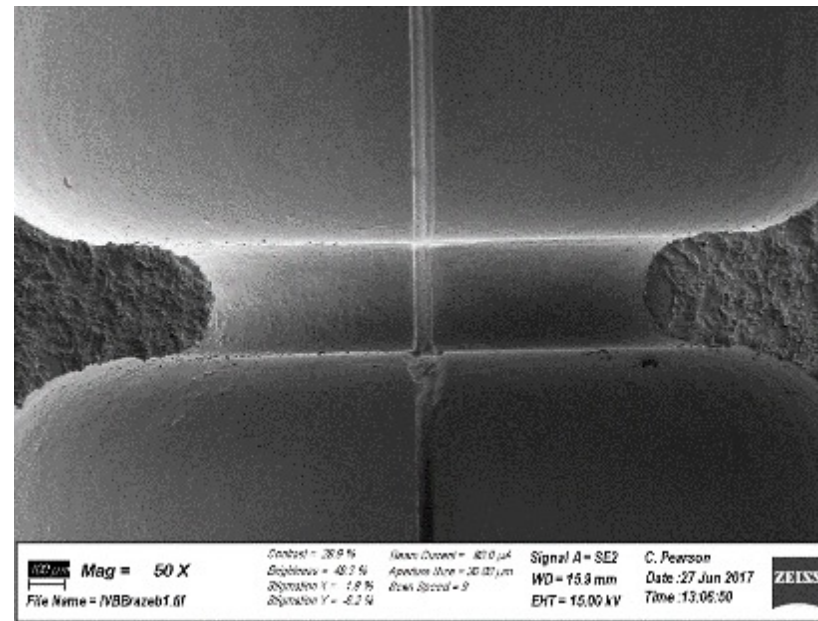


SRF: <https://sites.google.com/view/uspas2022-rf-superconductivity/home>

NCRF: <https://www.osti.gov/servlets/purl/1441166>

Other considerations

- Material preparation:
 - Contaminants, surface features
 - Especially important for SRF
 - Also a vacuum consideration
- Surfaces/joints:
 - Sharp edges cause breakdown
 - Improved by “processing” structure with successively higher input power
- Mechanical properties:
 - Heating (related to surface H field)
 - Vibrations, especially important for SRF (high Q, possible detuning)



SRF: <https://sites.google.com/view/uspas2022-rf-superconductivity/home>

NCRF: <https://www.osti.gov/servlets/purl/1441166>

Cavity tuning: mechanical

- Frequency set by boundary conditions, size of cavity features
- Can analyze effect of small perturbation, deformation of volume and surface, ΔV and ΔS

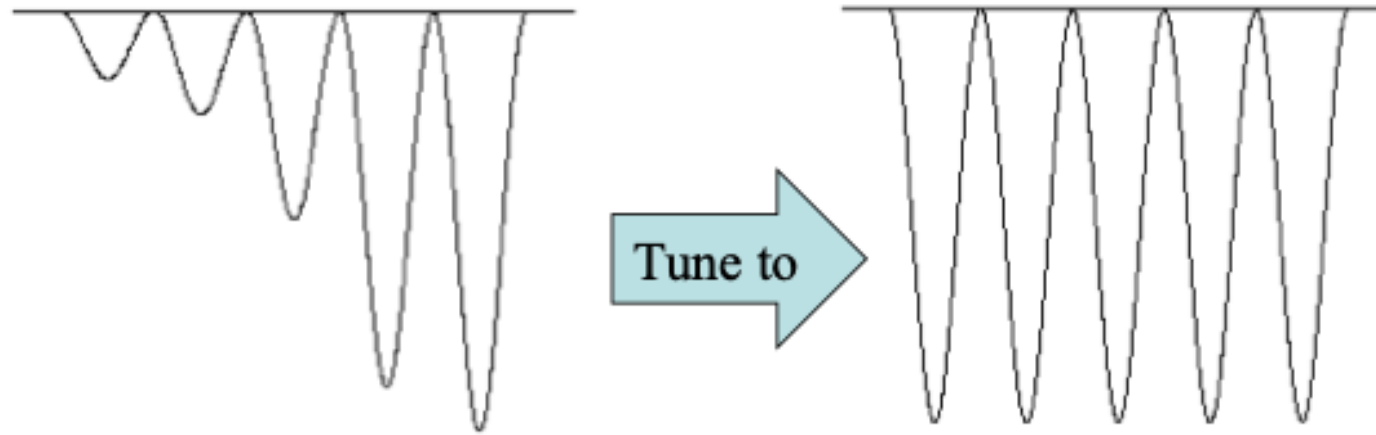
$$\frac{\omega - \omega_0}{\omega_0} \approx \frac{\int_{\Delta V} (\mu |\mathbf{H}_0|^2 - \epsilon |\mathbf{E}_0|^2) dv}{\int_{V_0} (\mu |\mathbf{H}_0|^2 + \epsilon |\mathbf{E}_0|^2) dv}.$$

$$\frac{\omega - \omega_0}{\omega_0} = \frac{\Delta U_m - \Delta U_e}{U_m + U_e}$$

- Assumes small perturbation
- Related to the change in electric and magnetic stored energy
- Can be temporary or permanent deformation

Cavity tuning: mechanical

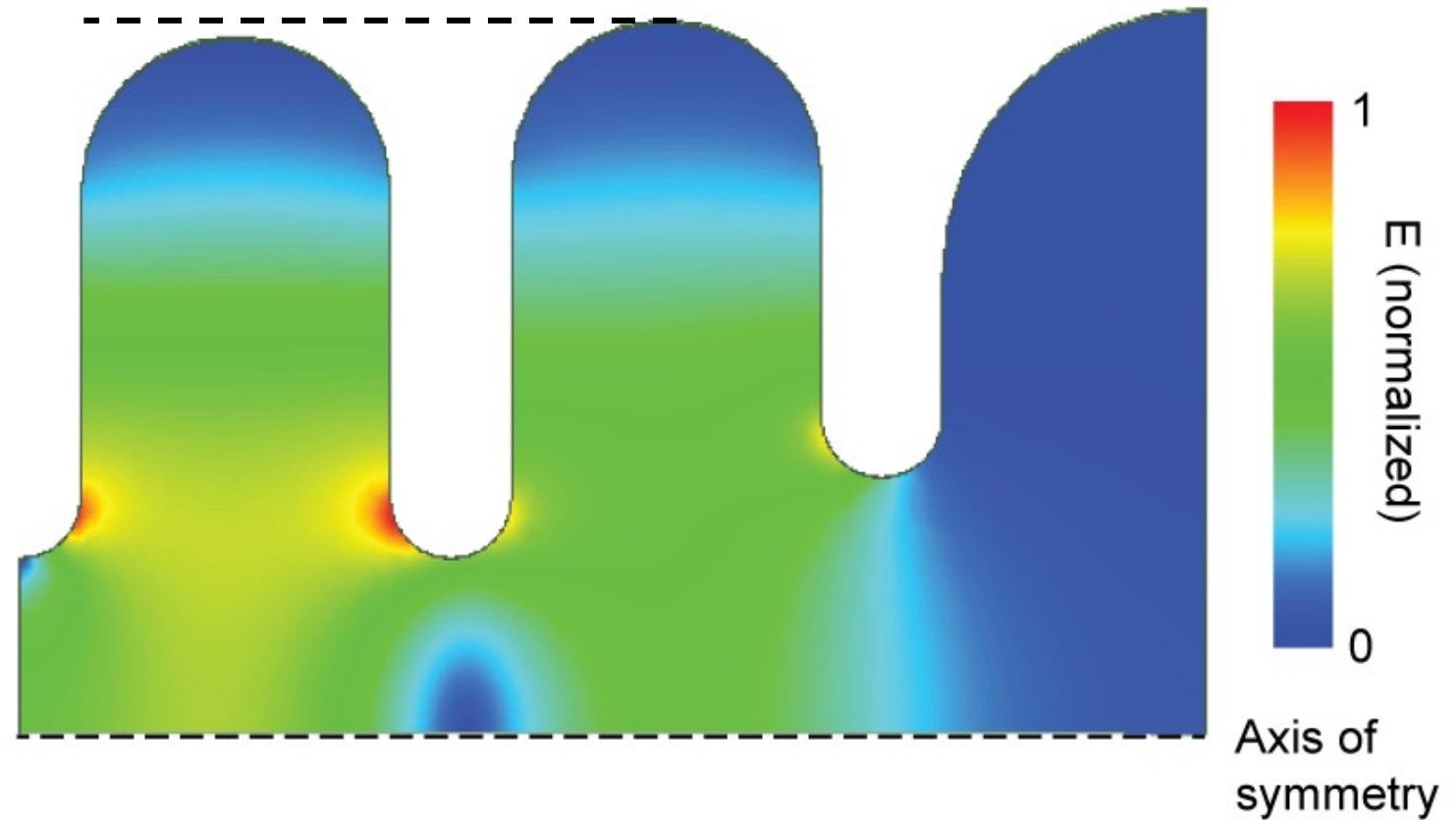
- Can tune one cell compared to another to shift field balance
- Most of the time, want a flat profile



<https://sites.google.com/view/uspas2022-rf-superconductivity/home>

Cavity tuning: mechanical

- Can tune one cell compared to another to shift field balance
- Most of the time, want a flat profile
- In other cases, may want localized field
- Example for an electron gun:



Cavity tuning: thermal tuning

- Metals undergo thermal expansion/contraction, depends on initial size
- Linear thermal expansion:

$$\frac{\Delta L}{L_0} = \alpha \Delta T$$

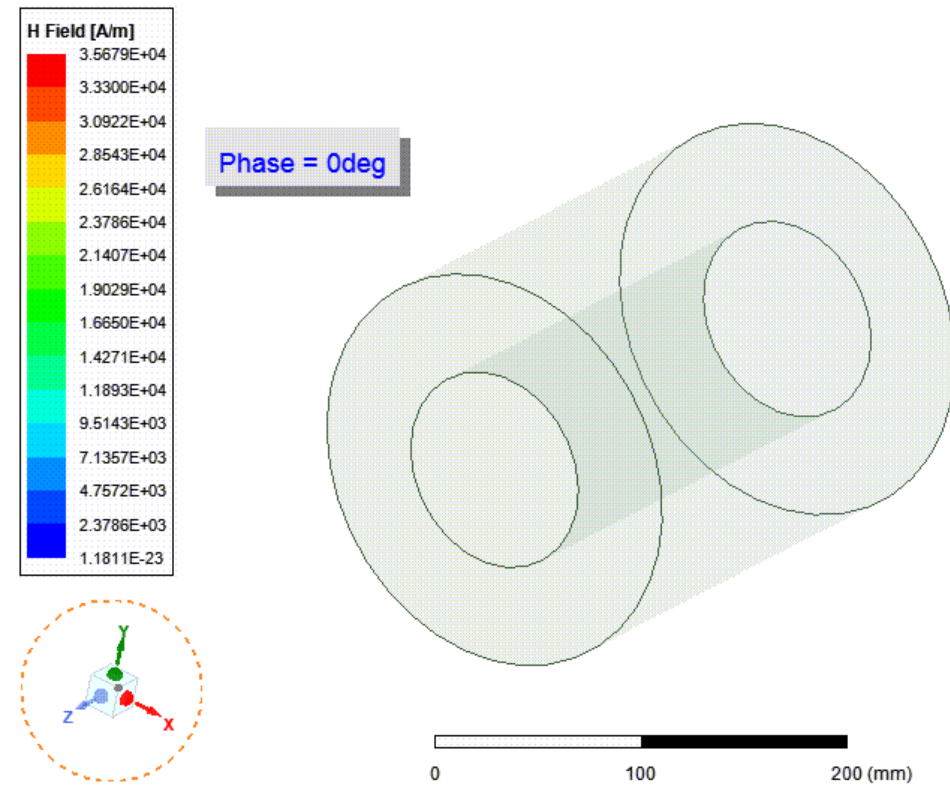
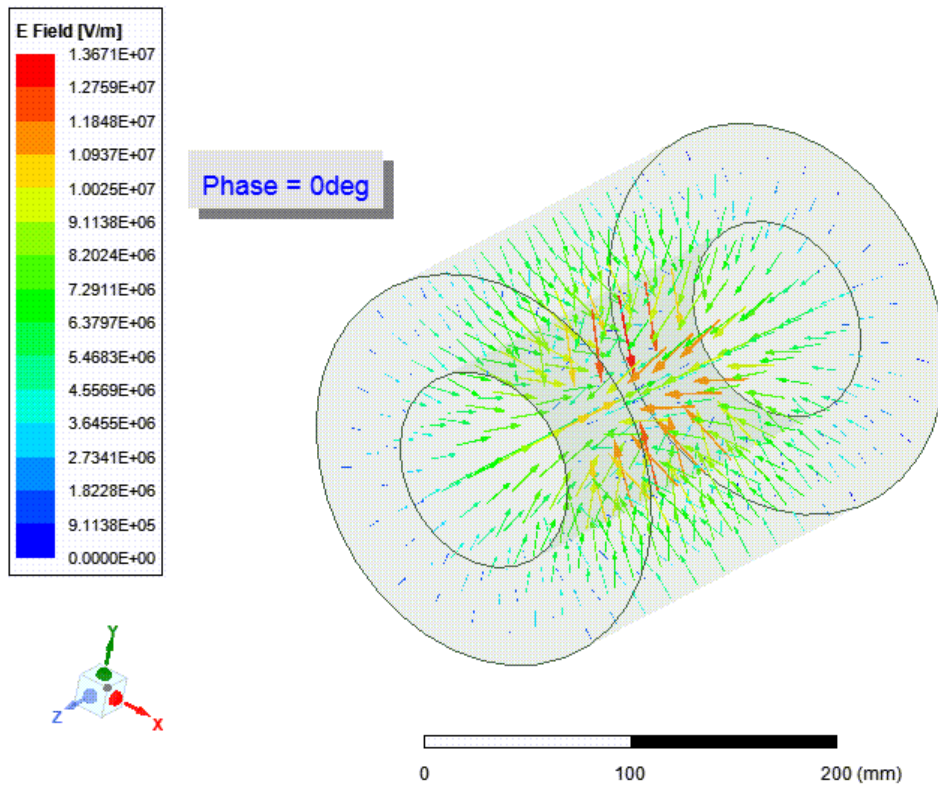
 Coefficient of thermal expansion

- Changing temperature changes size of cavity features, shifting frequency
- Overall size change rather than localized mechanical perturbation
- Useful for active tuning of normal conducting structures
- Note: frequency difference at room temperature vs cryogenic temperatures is important for SRF cavity design

More types of cavities

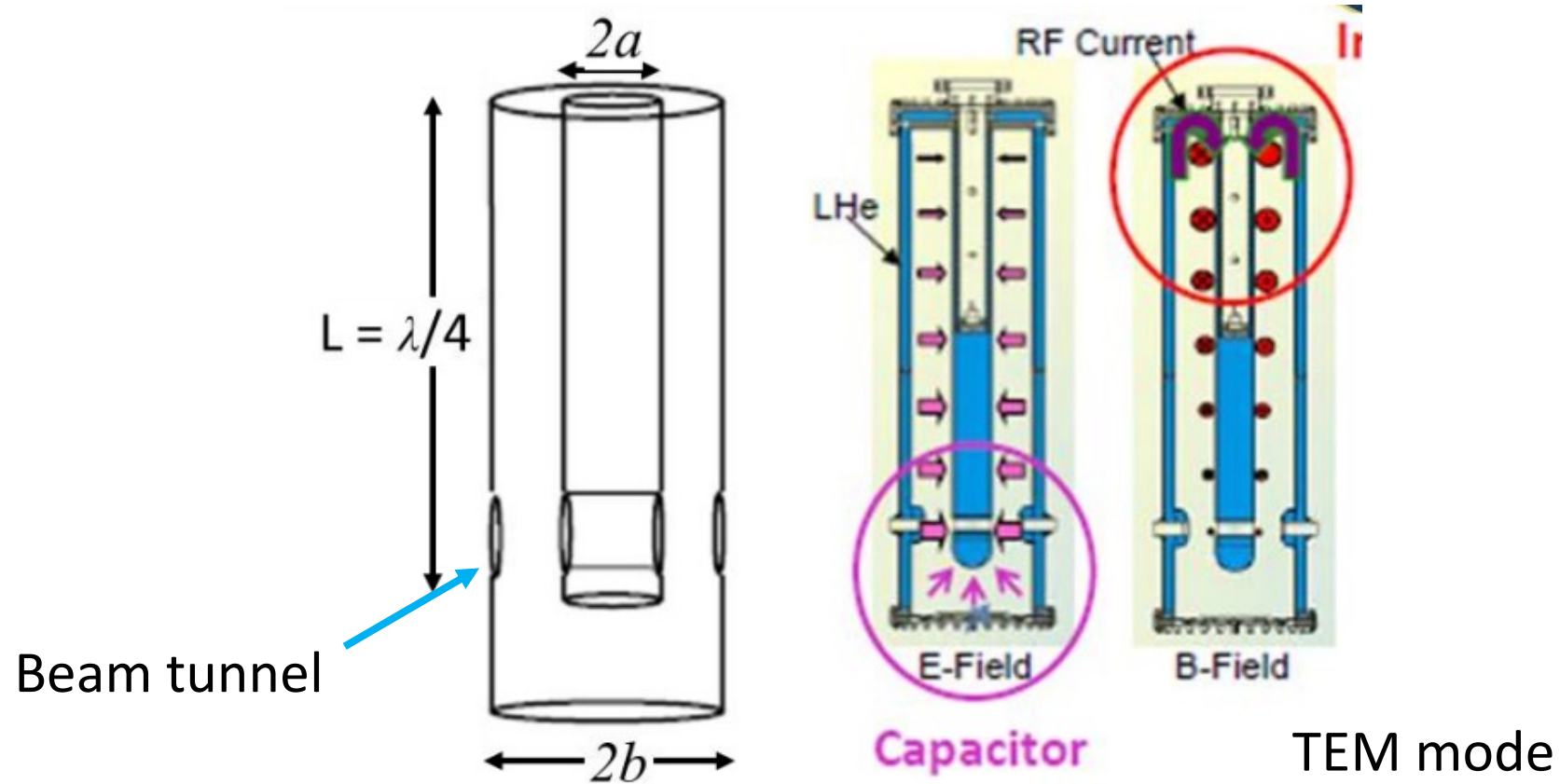
Low β cavity designs

- Can be impractical to shorten elliptical-style cavity for low particle velocity
- Most low β designs look more like coaxial transmission line, use TEM modes



Low β cavity designs

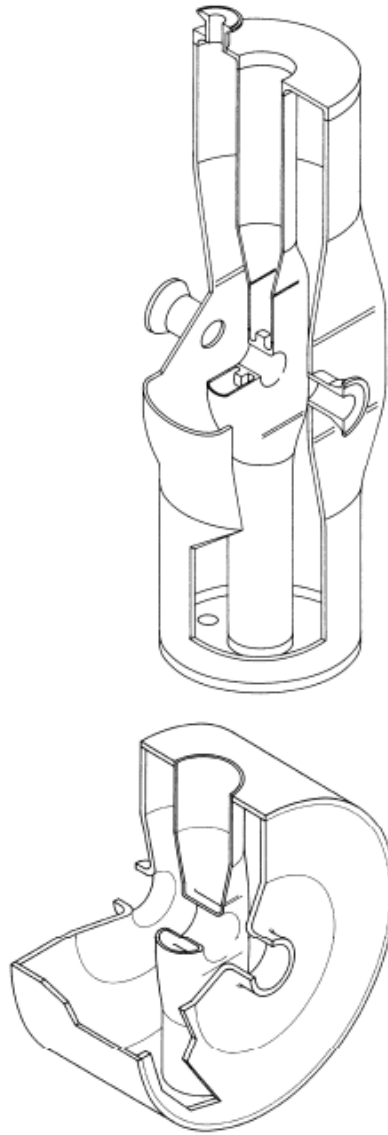
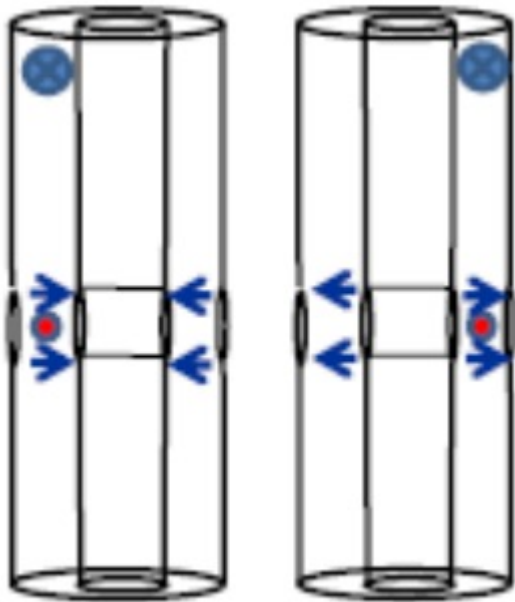
- Example: quarter wave resonator



https://casa.jlab.org/publications/viewgraphs/USPAS2020/L_10.pdf

Low β cavity design

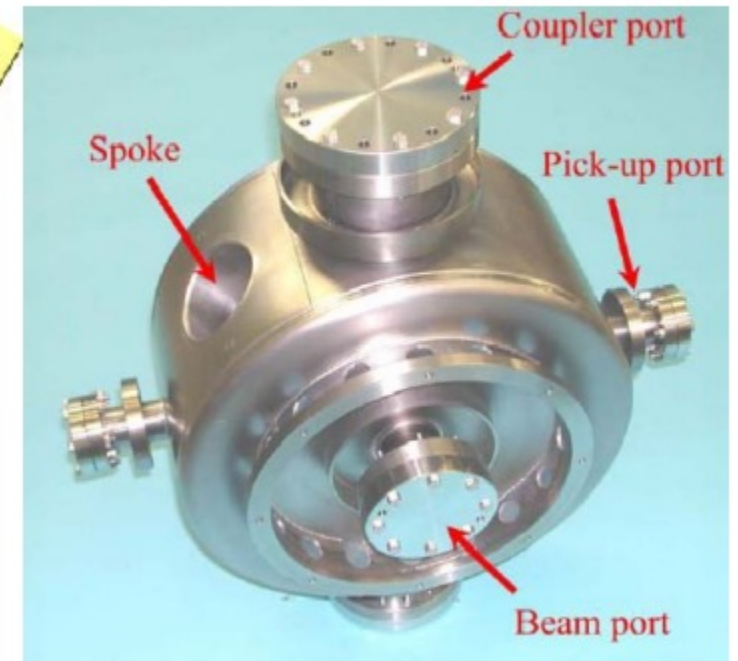
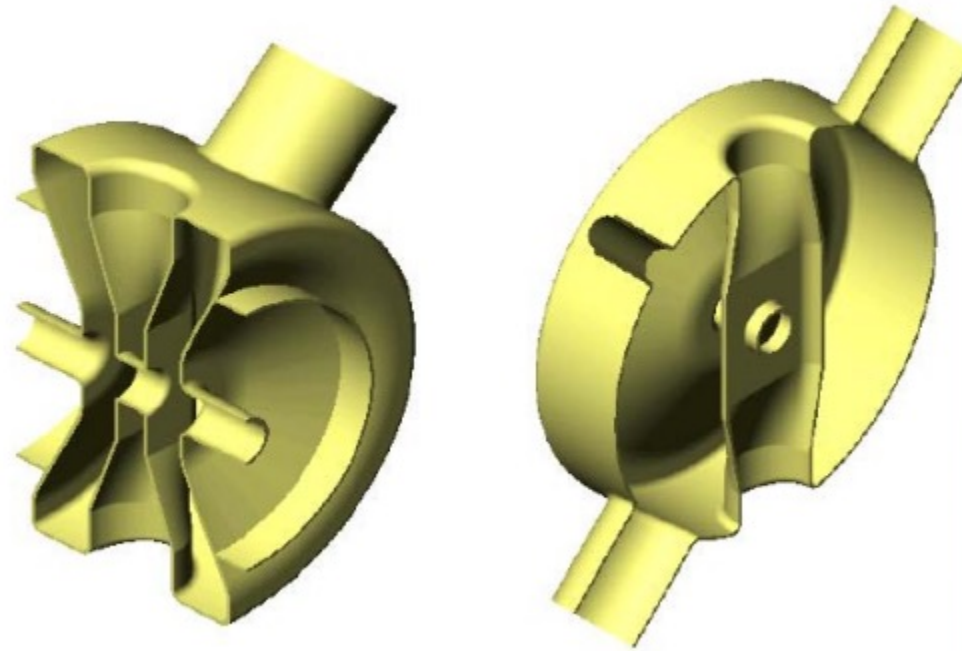
- Example: half wave resonator



https://casa.jlab.org/publications/viewgraphs/USPAS2020/L_10.pdf

Velocities in waveguides

- Example: spoke style half wave



https://casa.jlab.org/publications/viewgraphs/USPAS2020/L_10.pdf